INTERNSHIP MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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(2116240701357)

in partial fulfilment of the End Semester Examination for the course

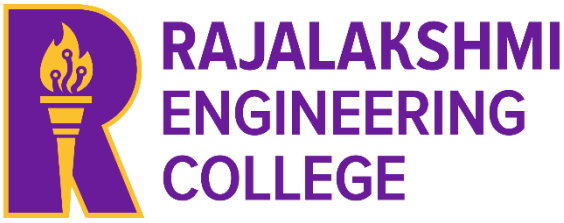
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

NOVEMBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "INTERNSHIP MANAGEMENT SYSTEM " is the bonafide work of NETHRA CHANDRAGANDHI T (2116240701357), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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Submitted for the End Semester Examination (Individual Project) held on 11 November 2026

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, NETHRA CHANDRAGANDHI T (2116240701357), hereby declare that the project report titled “Internship Management System”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 11 November 2026

SIGNATURE OF THE STUDENT
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I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

NETHRA CHANDRAGANDHI T

ABSTRACT

Internship coordination in an engineering college involves student profiles, company details, applications, attendance and performance evaluation. When these are kept in separate spreadsheets, registers and e-mails, the status of an application is hard to track, identifiers, durations and working hours are calculated by hand, and a consolidated view of the internship programme is slow to obtain.

This project presents the Internship Management System, a low-code application developed on Zoho Creator. Five linked forms manage student registration, company details, internship applications, attendance and evaluation. Seven Deluge workflows generate application IDs, calculate internship duration, attendance working hours, performance result and attendance percentage, update approval and application status, and send an approval notification e-mail. An approval process, a Blueprint with the stages Applied, Under Review, Selected and Rejected, and two schedules support the application life cycle. Thirteen reports (list, spreadsheet, Kanban, calendar, timeline, map and pivot views) and a dashboard with record counts, a status pie chart, a working-hours bar chart and an average-rating gauge present the stored data. An Admin role and Student and Customer portal permission profiles are configured.

The application was tested in the Development environment. Application IDs were generated automatically, an internship from 1 November 2026 to 1 December 2026 gave 30 days, a 09:00 to 15:00 attendance record gave 6.00 working hours, ratings of 5, 4, 5, 4 and 5 gave an overall rating of 4.6, and the approval e-mail was received at the configured test address. Delivery to each student's own e-mail address, role-based testing and deployment beyond the Development environment remain as future work. The project shows that a low-code platform can provide a working internship record-management solution with limited scripting.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Approval Process; Internship Management*

PROJECT AT A GLANCE

Attribute	Details
Project Title	Internship Management System using Zoho Creator
Application Domain	Education – internship management
Student Name & Register No.	NETHRA CHANDRAGANDHI T – 240701357
Zoho Creator Edition	Trial edition (as shown in the application header)
Application Name (Link Name)	Internship Management System (internship-management-system)
Published / Live URL	https://creatorapp.zoho.in/nethrachandragandhi.t.202102/environment/development/internship-management-system (Development environment)
Demo Credentials (per role)	Admin: nethrachandragandhi.t.2024.cse@rajalakshmi.edu.in
Number of Forms /	5 / 0
Number of Reports (by type)	List: 6 Spreadsheet: 1 Kanban: 1 Calendar: 1 Timeline: 1 Map: 1 Pivot/Chart: 2 (Total: 13)
Number of Pages / Dashboards	1 dashboard (Internship Dashboard)
Number of Workflows	Form: 7 Schedule: 2 Approval: 1 Blueprint: 1 Batch: 0
Deluge Functions / Custom Actions	7 Deluge scripts in form workflows / 0 custom functions
Integrations	None external; built-in Send Mail action and Map report
Roles & Profiles Defined	Roles: CEO, Admin | Portal profiles: Customer, Student
Environments Used	Development
Date of Production Deployment	25 October 2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CGPA	Cumulative Grade Point Average
CSE	Computer Science and Engineering
DS	Deluge Script (script identifier)
ER	Entity–Relationship
FR	Functional Requirement
HoD	Head of the Department
NFR	Non-Functional Requirement
REST	Representational State Transfer
TC	Test Case
UAT	User Acceptance Testing
URL	Uniform Resource Locator

CHAPTER 1
INTRODUCTION

1.1 Background

Internships are an important part of engineering education because they give students exposure to real industry practices. Managing them in an institution involves several kinds of information: the profiles of the students, the companies offering internships, the applications submitted by students, the daily attendance of interns and the evaluation of their performance. When these records are kept in separate spreadsheets, registers and e-mail threads, coordinators spend considerable effort in entering, cross-checking and consolidating data.

This project addresses that domain through the **Internship Management System**, a low-code application developed on **Zoho Creator** [2]. The application is designed for an internship coordination office of an educational institution. It brings student registration, company details, internship applications, attendance tracking and performance evaluation into one application, and adds workflows, reports and a dashboard on top of the stored data. The application is built and tested in the Development environment of the student's Zoho Creator account.

1.2 Problem Statement

Internship-related information is commonly maintained in scattered files and registers, which makes it difficult to identify an application's status, to calculate durations and working hours consistently, and to obtain a consolidated view of students, companies, attendance and evaluation results. There is a need for a single, structured application that stores this information, automates the routine calculations and presents it through reports and a dashboard.

- Student, company and application records are kept in different places and are not linked to each other.
- Application identifiers, internship durations and attendance working hours are calculated or typed manually, which can lead to errors.
- The progress of an application (applied, under review, selected or rejected) and its approval are not tracked in a structured manner.

- A consolidated view of record counts, application status, working hours and evaluation ratings is not readily available.

1.3 Objectives

- To design forms and a linked data model for student registration, company details, internship applications, attendance and evaluation.
- To automate application ID generation, internship duration, attendance working hours, performance result and attendance percentage using Deluge scripting in form workflows.
- To control the application life cycle using an approval process, a Blueprint and a field-driven approval workflow, with an e-mail notification on approval.
- To provide reports in multiple views (list, spreadsheet, Kanban, calendar, timeline, map and pivot) and a dashboard with key metrics and charts.
- To configure an Admin role and a Student portal permission profile for role-based access.
- To test the forms, workflows, reports, dashboard and e-mail notification in the Development environment.

1.4 Scope of the Project

In scope: five forms (Student Registration, Company Details, Internship Application, Internship Attendance and Internship Evaluation); seven form workflows written in Deluge; one approval process; one Blueprint; two scheduled workflows; thirteen reports; one dashboard; an Admin role and Student and Customer portal permission profiles; and testing of these components in the Development environment.

Out of scope: dynamic e-mail delivery to each student's registered address (the tested notification uses a fixed test address), certificate management, multi-level approvals, integration with external systems or APIs, payment handling, and pro-code extensions. Use of the Zoho Creator mobile app and a live customer-portal login were not evaluated in this project.

1.5 Justification for a Low-Code Approach

Low-code platforms allow business applications to be assembled mainly through visual builders, with scripting used only where logic cannot be configured [1]. This suits the present application because most of its requirements (data-entry forms, linked records,

reports, a dashboard, approvals and scheduled mail) are provided as built-in components of Zoho Creator [2], while the calculations that need logic (ID generation, durations, working hours and ratings) are small and are handled in Deluge scripts [3]. Hosting, the user interface and role-based access are provided by the platform, so the effort is concentrated on the internship process itself rather than on infrastructure.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In a typical manual or spreadsheet-based arrangement, a student collects details of a company, submits an application on paper or by e-mail, and the coordinator records it in a spreadsheet. Approval is communicated through phone calls or e-mail. During the internship, attendance is noted in a register and performance is recorded on paper forms. Reports are finally prepared by combining these separate sources. Figure 2.1 shows this typical process.

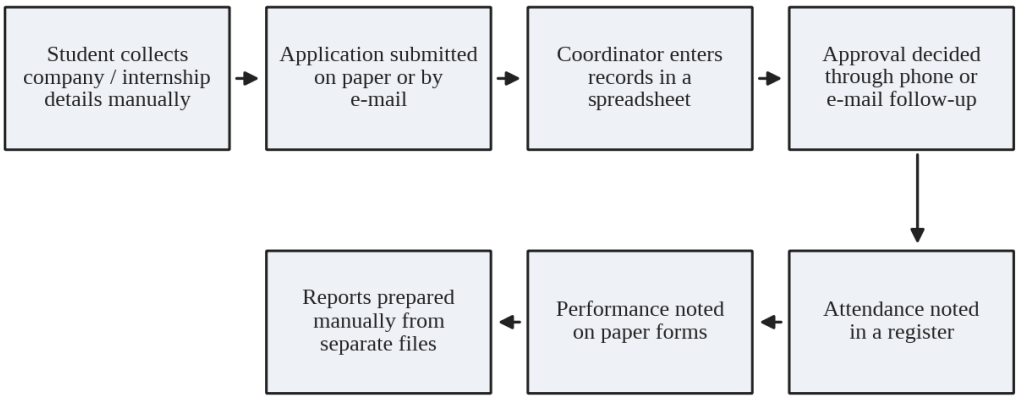


Figure 2.1: Typical Manual (As-Is) Process for Managing Internships

2.2 Limitations of the Existing System

- Records of students, companies, applications, attendance and evaluations are stored separately and are not linked.
- Application numbers, internship durations and working hours are entered or calculated manually.
- The status of an application and its approval are not tracked through a defined process.
- Reports must be compiled by hand, so a consolidated view is slow to obtain.
- Notifications about approval depend on the coordinator remembering to inform the student.

2.3 Low-Code versus Traditional Development

Table 2.1 compares traditional development with the low-code approach for this application. The comparison is qualitative; no formal effort measurements were taken during the project.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development effort	Database design, back-end code, user interface and report screens must all be programmed.	Forms, reports and the dashboard are assembled in visual builders; Deluge is used for seven small scripts.
Skills required	Programming, database and web-development skills.	Platform familiarity and basic Deluge scripting.
Hosting and infrastructure	Servers, deployment and maintenance are managed by the developer.	Hosted by Zoho; the application is accessed through a web link.
Workflow and approvals	Approval logic and scheduling must be coded.	Approval process, Blueprint and schedules are configured in the Workflow builder.
Security and access	Authentication and authorisation must be designed and coded.	Roles and portal permission profiles are provided by the platform.
Customisation limits	Unlimited flexibility.	Bounded by platform features; Deluge covers most custom logic needed here.
Cost	Developer time, server and licence costs.	Developed on a Zoho Creator trial edition; a subscription would be needed for continued use.
Maintenance	Code changes, testing and redeployment.	Changes are made in the builders or in Deluge and tested in the Development environment.

2.4 Literature Survey

The sources reviewed for this project are the official Zoho Creator documentation, the Deluge reference, the Zoho Creator learning material and a textbook on low-code development. They are summarised in Table 2.2.

Table 2.2: Summary of Literature Survey

S.No.	Author / Source & Year	Title	Key Findings	Relevance to this Project
1	Zoho Corporation, online documentation 30 October 2026	Zoho Creator Help Documentation [2]	Describes the form builder and field types, report types, workflows, approvals, Blueprint, schedules, roles and pages.	Primary reference for building the five forms, reports, workflows and the dashboard.
2	Zoho Corporation, online documentation 30 October 2026	Deluge Script Reference Guide [3]	Describes Deluge syntax, conditions, date and time handling, fetching records, collections and the sendmail task.	Basis for the Deluge scripts DS-01 to DS-07.
3	Zoho Corporation, online learning material 30 October 2026	Zoho Creator Learning Centre [4]	Provides learning material on platform features.	Supporting material for features used in the application.
4	W. Rosam and O. Kalnichevski	Low-Code Development: A Practical Guide [1]	Introduces low-code concepts and platforms.	Supports the low-code justification and the no-code to pro-code discussion.
5	Authors: D. Liu et al.	What's Wrong With Low-Code Development Platforms? An Empirical Study of Low-Code Development Platform Bugs	Identifies common bugs in low-code platforms, including application design issues, unexpected system behavior, and user interface graphics problems	Supports testing workflows, approval processes, attendance calculations, and reports for system reliability.

2.5 Emerging Trends in Low-Code Development

Low-code platforms are increasingly used by business teams to build applications without long development cycles, a practice often described as citizen development. Platforms such as Zoho Creator combine visual builders with scripting, built-in workflow automation, approvals and analytics, and now also provide AI-assisted building (the Zoho Creator editor used in this project displays a Zia “Build Agent” panel). These trends are relevant to this project because its core elements (forms, approvals, scheduled mail and dashboards) are the kind of components that low-code platforms offer out of the box.

2.6 Proposed System

The proposed Internship Management System stores student, company, application, attendance and evaluation data in five linked forms. Deluge workflows generate the

application ID and calculate the internship duration, working hours, performance result and attendance percentage. An approval process and a Blueprint control the application life cycle, a field-driven workflow updates the approval and application status and sends a notification e-mail, and scheduled workflows send reminders to the administrator. Reports in several views and a dashboard provide a consolidated picture. In this way the system addresses each limitation listed in Section 2.2.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Owner of the application; the Admin role is configured under the CEO role in the role hierarchy, and the owner account is the Level 1 approver of the approval process.	Maintain student and company records, create and review applications, approve or reject applications, record attendance and evaluations, view reports and the dashboard.
Student (Portal User)	Student profile defined in the portal user permissions: "allows students to view permitted internship information and manage their own applications".	Submit and manage own internship applications; view permitted internship information.
Company Mentor	Named in the Mentor Name field of the Internship Evaluation form; not configured as a system user.	Provides the ratings that are recorded in the evaluation.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall register a student with mandatory fields (ID, name, e-mail, phone, gender, department, year, CGPA, skills).	Admin	High
FR-02	The system shall store company profiles and internship opportunity details.	Admin	High
FR-03	The system shall record an internship application linked to a student and a company and generate an Application ID automatically on submission.	Admin, Student	High
FR-04	The system shall calculate the internship duration in days from the start and end dates.	Admin	High
FR-05	The system shall record attendance with check-in and check-out times and calculate working hours.	Admin	High
FR-06	The system shall record five performance ratings (1 to 5) and compute the overall rating as their average.	Admin	High
FR-07	The system shall derive a Performance Result from the overall rating.	Admin	Medium

FR ID	Requirement	Role(s)	Priority
FR-08	The system shall set Approval Status to Approved and Application Status to Selected when Application Approved is ticked.	Admin	High
FR-09	The system shall send an “Internship Application Approved” e-mail when the Application Approved field is updated.	Admin	Medium
FR-10	The system shall route each application through an approval process with Approve and Reject actions.	Admin	High
FR-11	The system shall move an application through the stages Applied, Under Review, Selected and Rejected using a Blueprint.	Admin	Medium
FR-12	The system shall calculate a student’s attendance percentage from attendance records.	Admin	Medium
FR-13	The system shall provide reports in list, spreadsheet, Kanban, calendar, timeline, map and pivot views.	Admin	Medium
FR-14	The system shall display a dashboard of record counts, application status, working hours and average evaluation rating.	Admin	Medium
FR-15	The system shall send scheduled reminder mails to the administrator.	Admin	Low
FR-16	The system shall restrict access using an Admin role and Student and Customer portal permission profiles.	Admin, Student	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports and the dashboard shall display the stored records without noticeable delay for the data volumes used in the project.
NFR-02	Security	Users shall access only the data permitted by their role or portal profile.
NFR-03	Usability	The application shall be usable through a web browser with clearly labelled forms and mandatory-field indicators.
NFR-04	Availability	The application shall be hosted on the Zoho Creator platform and accessed through a web link.
NFR-05	Maintainability	Logic shall be kept in small, separately named workflows so that each rule can be changed independently.
NFR-06	Data integrity	Essential fields shall be mandatory, and calculated values shall be produced by workflows rather than typed manually.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (trial edition, as displayed in the application header)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome (used for development and testing); other current browsers are expected to work
Other Zoho services	Send Mail action (notification e-mail); Map report (Zoho Maps)
Mobile	Zoho Creator mobile app: not evaluated in this project
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

Figure 3.1 shows the main use cases for the Administrator and the Student (Portal User). The Student's use cases follow the description of the "Student" portal permission profile. Table 3.5 describes the main use case in detail.

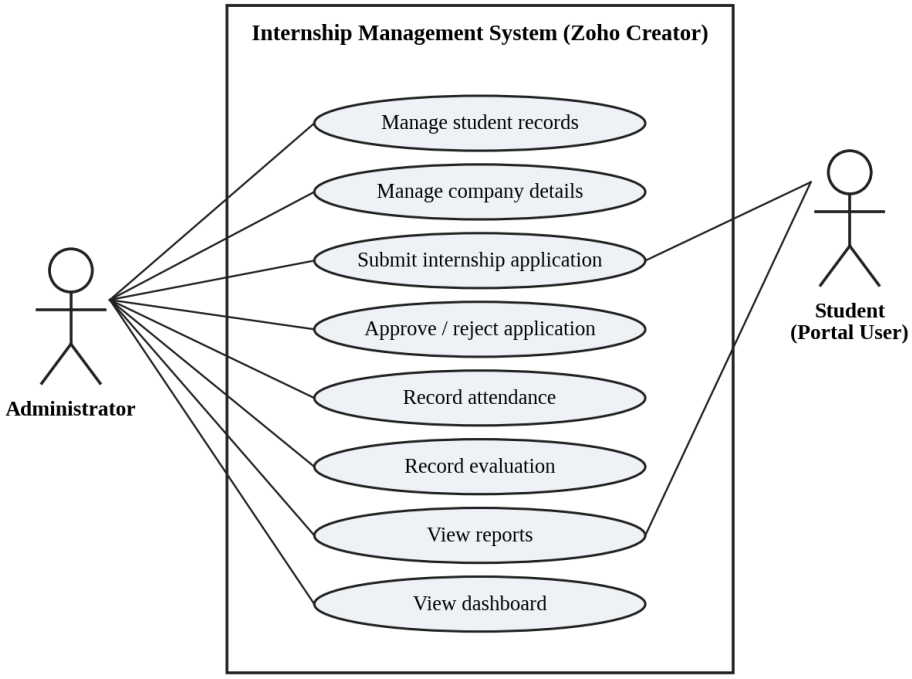


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Submit and Approve an Internship Application

Use Case ID	UC-01
Actors	Student; Administrator

Preconditions	The student and the company already exist in the Student Registration and Company Details forms.
Main Flow	1. The student submits an internship application (student, company, domain, dates, mode). 2. DS-01 generates the Application ID. 3. The record is saved and an approval request appears in Pending Tasks for the approver. 4. The administrator reviews the application and ticks Application Approved. 5. DS-03 sets Approval Status to Approved and Application Status to Selected; DS-02 calculates the duration. 6. The approval notification e-mail is sent.
Alternate Flow	The approver rejects the request in the approval process, or the Blueprint transition “Reject Application” moves the application to Rejected.
Postconditions	The application shows its updated status in the reports and dashboard.
Related Requirements	FR-03, FR-04, FR-08, FR-09, FR-10, FR-11

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

Figure 4.1 shows the layered architecture. Users (the Administrator and the Student portal user) reach the application through a web browser. The application layer holds the forms, reports and dashboard together with the workflows (form workflows written in Deluge, the approval process, the Blueprint and the schedules). Data is stored in five Zoho Creator forms. The platform provides the Send Mail action used for notifications and the Zoho Maps service used by the Map report.

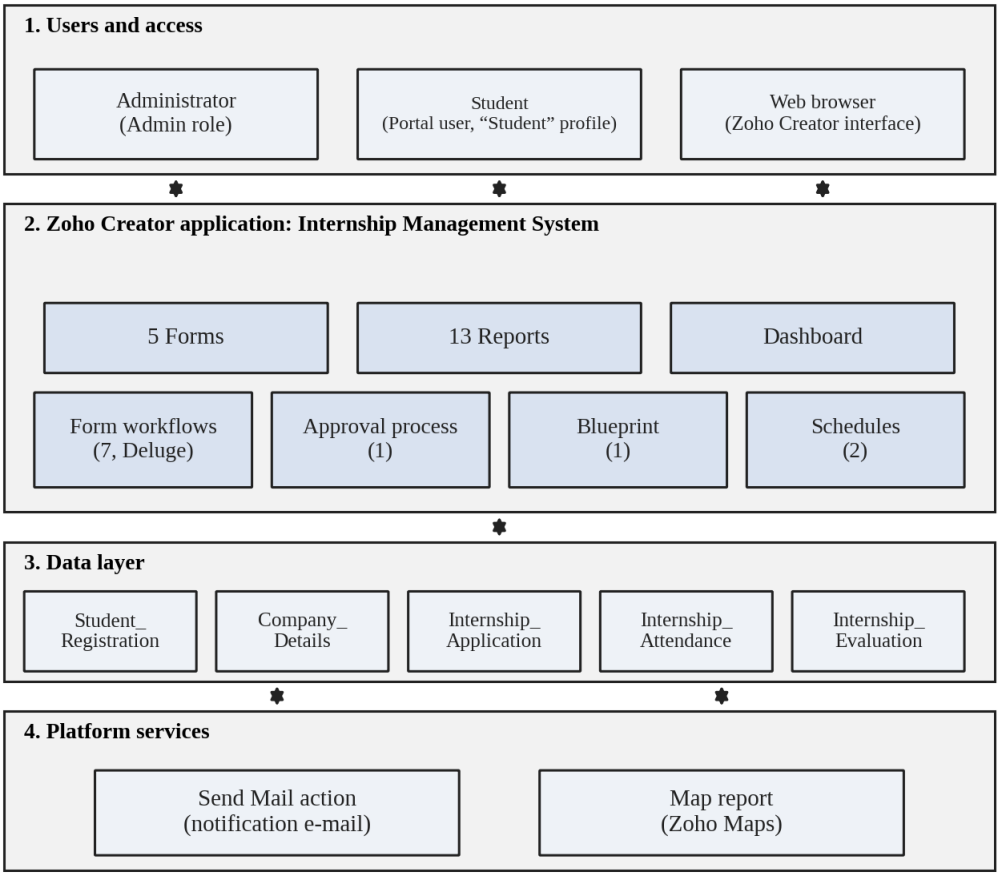


Figure 4.1: System Architecture of the Application

4.2 Data Model

Figure 4.2 presents the entity–relationship diagram. Student Registration and Company Details are the master forms. Internship Application, Internship Attendance and Internship Evaluation each refer to a student and a company through lookup fields, so

one student or company can be related to many applications, attendance records and evaluations. No subforms are used. Table 4.1 lists the entities and relationships.

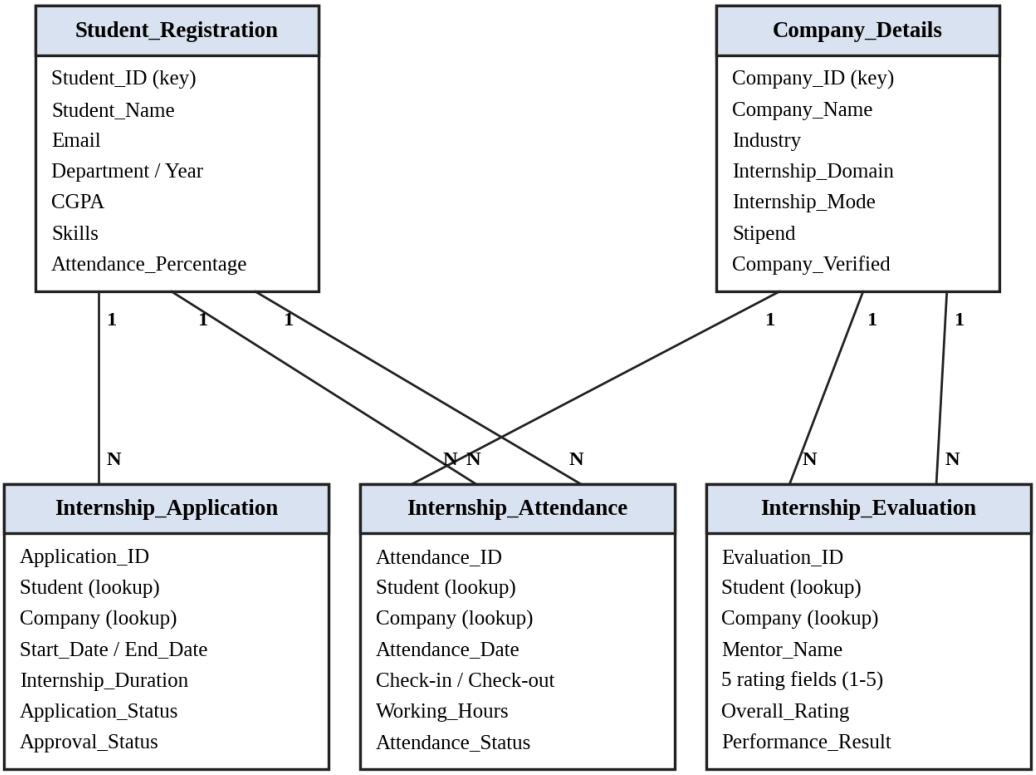


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship	Implemented By
Student Registration	Student ID, Student Name, Email, Department, Year, CGPA, Skills	Application, Attendance, Evaluation	One student to many records	Lookup fields in the three child forms
Company Details	Company ID, Company Name, Industry, Internship Domain, Mode, Duration	Application, Attendance, Evaluation	One company to many records	Lookup fields in the three child forms
Internship Application	Application ID, Start Date, End Date, Application Status, Approval Status	Student Registration, Company Details	Many applications to one student and one company	Student Registration and Company Details lookups
Internship Attendance	Attendance ID, Attendance Date, Check-In/Out Time, Working Hours	Student, Company	Many records to one student and one company	Student and Company lookups

Entity (Form)	Key Fields	Related To	Relationship	Implemented By
Internship Evaluation	Evaluation ID, five ratings, Overall Rating, Performance Result	Student, Company	Many records to one student and one company	Student and Company lookups

4.3 Form Design

Each form is specified below. Link names are those generated by Zoho Creator from the field labels; the field types, mandatory markers and values are taken from the live forms. The forms together use more than ten distinct field types: single line, name, e-mail, phone, radio, date, dropdown, decimal, number, multi-select, file upload, image, multi line, decision box, URL, address, currency, time, lookup and formula.

Table 4.2: Field Specification – Student Registration Form

Field Label	Link Name	Field Type	Mand.	Validation / Values	Remarks
Student ID	Student_ID	Single Line	Yes	—	Identifier, e.g. STU001
Student Name	Student_Name	Name	Yes	First and last name	
Email	Email	Email	Yes	E-mail format	
Phone Number	Phone_Number	Phone	Yes	Country code (+91 shown)	
Gender	Gender	Radio	Yes	Male / Female	
Date of Birth	Date_of_Birth	Date	No	dd-MMM-yyyy	
Department	Department	Dropdown	Yes	Choice list	e.g. CSE, IT
Year	Year	Dropdown	Yes	Choice list	e.g. 3rd year
Semester	Semester	Dropdown	No	Choice list	
CGPA	CGPA	Decimal	Yes	Decimal number	
Attendance Percentage	Attendance_Percentage	Number	No	Set by DS-07	
Skills	Skills	Multi Select	Yes	Choice list	
Resume	Resume	File Upload	No		
Profile Photo	Profile_Photo	Image	No		
Address	Address	Multi Line	No		
Placement Interested	Placement_Interested	Decision Box	No	Checkbox	

Table 4.3: Field Specification – Company Details Form

Field Label	Link Name	Field Type	Mand.	Validation / Values	Remarks
Company ID	Company_ID	Single Line	Yes	—	e.g. COM001
Company Name	Company_Name	Single Line	Yes		
Company Logo	Company_Logo	Image	No		
Industry	Industry	Dropdown	Yes	Choice list	e.g. Software, IT
Contact Person	Contact_Person	Name	Yes	First and last name	
Company Email	Company_Email	Email	Yes	E-mail format	
Contact Number	Contact_Number	Phone	Yes	Country code (+91 shown)	
Company Website	Company_Website	URL	No	https://	
Company Address	Company_Address	Address	Yes	Address lines, city, state, postal code, country	Used by the Map report
Internship Domain	Internship_Domain	Multi Select	No	Choice list	e.g. AI/ML
Internship Mode	Internship_Mode	Radio	No	On-site / Hybrid / Remote	
Internship Duration	Internship_Duration	Dropdown	Yes	Choice list	e.g. 2 Months
Stipend	Stipend	Currency	No		
Available Positions	Available_Positions	Number	Yes		
Company Description	Company_Description	Multi Line	No		
Company Verified	Company_Verified	Decision Box	No	Checkbox	

Table 4.4: Field Specification – Internship Application Form

Field Label	Link Name	Field Type	Mand.	Validation / Values	Remarks
Application ID	Application_ID	Single Line	No	Generated by DS-01	Format APP-yyyyMMddHHmmss
Student Registration	Student_Registration	Lookup	No	Student Registration form	
Company Details	Company_Details	Lookup	No	Company Details form	

Field Label	Link Name	Field Type	Mand.	Validation / Values	Remarks
Internship Domain	Internship_Domain	Dropdown	Yes	Choice list	e.g. AI/ML, Data Analytics
Application Date	Application_Date	Date	Yes	dd-MMM-yyyy	
Start Date	Start_Date	Date	Yes	dd-MMM-yyyy	
End Date	End_Date	Date	Yes	dd-MMM-yyyy	
Internship_Duration	Internship_Duration	Number	No	Days; set by DS-02	
Internship Mode	Internship_Mode	Radio	Yes	On-site / Hybrid / Remote	
Cover Letter	Cover_Letter	Multi Line	No		
Resume	Resume	File Upload	No		
Application Status	Application_Status	Dropdown	Yes	Applied, Under Review, Selected, Rejected, Withdrawn, Completed	
Approval Status	Approval_Status	Dropdown	Yes	Pending, Approved, Rejected	
Application Approved	Application_Approved	Decision Box	No	Checkbox	Triggers DS-03, DS-04
Admin Remarks	Admin_Remarks	Multi Line	No		

Table 4.5: Field Specification – Internship Attendance Form

Field Label	Link Name	Field Type	Mand.	Validation / Values	Remarks
Attendance ID	Attendance_ID	Single Line	Yes	—	e.g. ATT001
Student	Student	Lookup	Yes	Student Registration form	
Company	Company	Lookup	Yes	Company Details form	
Attendance Date	Attendance_Date	Date	Yes	dd-MMM-yyyy	
Attendance Status	Attendance_Status	Radio	Yes	Present, Absent, Leave, Half Day	
Check-In Time	Check_In_Time	Time	No	HH:mm:ss	
Check-Out Time	Check_Out_Time	Time	No	HH:mm:ss	
Working Hours	Working_Hours	Decimal	No	Set by DS-05	Two decimal places

Field Label	Link Name	Field Type	Mand.	Validation / Values	Remarks
Proof / Document	Proof_Document	File Upload	No		
Remarks	Remarks	Multi Line	No		

Table 4.6: Field Specification – Internship Evaluation Form

Field Label	Link Name	Field Type	Mand.	Validation / Values	Remarks
Evaluation ID	Evaluation_ID	Single Line	Yes	—	e.g. EVAL001
Student	Student	Lookup	Yes	Student Registration form	
Company	Company	Lookup	Yes	Company Details form	
Mentor Name	Mentor_Name	Name	Yes	First and last name	
Technical Skills	Technical_Skills	Number	Yes	Rating 1 to 5	Default 1
Communication	Communication	Number	Yes	Rating 1 to 5	Default 1
Teamwork	Teamwork	Number	Yes	Rating 1 to 5	Default 1
Problem Solving	Problem_Solving	Number	Yes	Rating 1 to 5	Default 1
Punctuality	Punctuality	Number	Yes	Rating 1 to 5	Default 1
Overall Rating	Overall_Rating	Formula	—	Average of the five ratings	Shown in the report
Performance Result	Performance_Result	Single Line	No	Set by DS-06	

4.4 Process and Workflow Design

Figure 4.3 shows the process flow for an internship application. Part A covers submission and the approval request, Part B the stages defined in the Blueprint, and Part C the actions executed when the Application Approved field is ticked. The approval process and the Application Approved workflow are separate configurations and are described separately in Section 5.6.

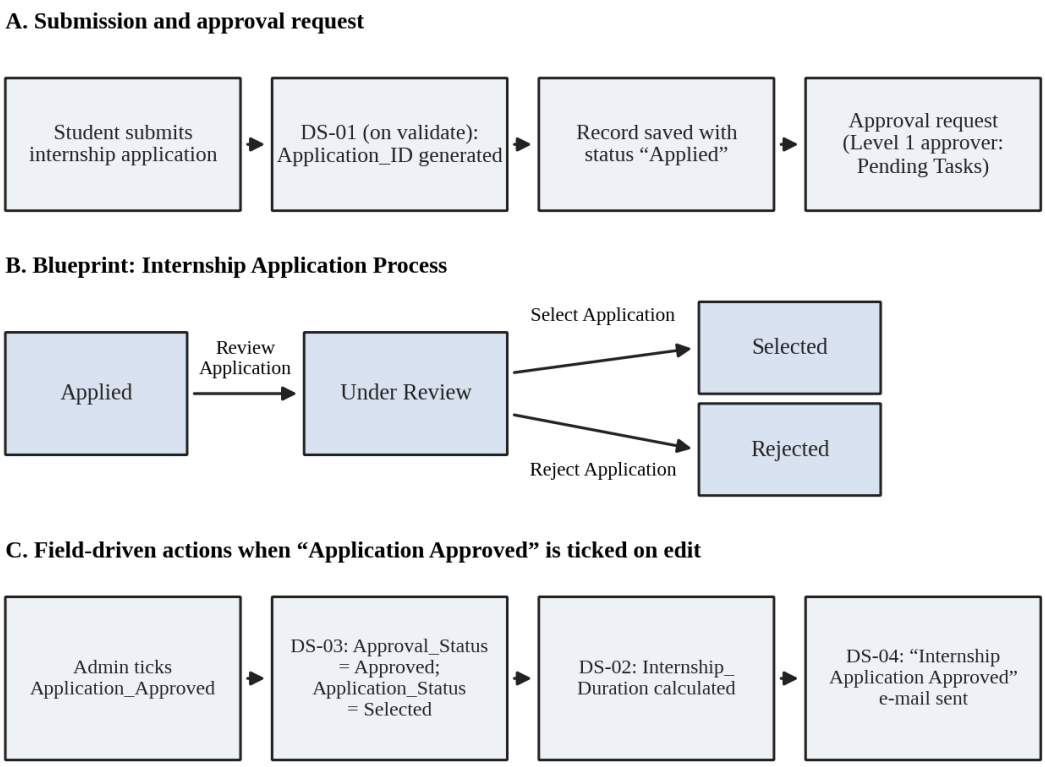


Figure 4.3: Workflow / Process Flow for the Internship Application

4.5 User Interface and Navigation Design

The application menu is organised by module, as shown in Figure 4.4. Each module groups its form with its reports, and the Internship Dashboard is a separate top-level entry.

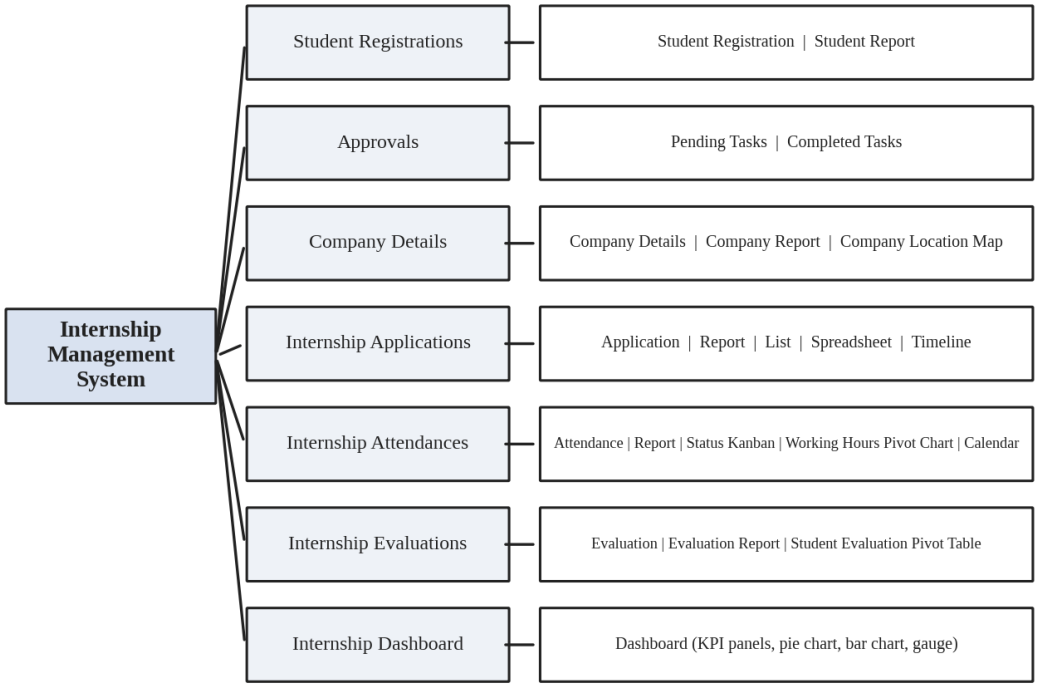


Figure 4.4: Navigation Structure of the Application

4.6 Access Control Design

Table 4.7 summarises the intended access design. Two access mechanisms are defined in the application: a role hierarchy for application users (CEO and Admin) and portal user permission profiles (Customer and Student). The Student column follows the description of the "Student" profile (view permitted internship information; manage own applications); the component-level settings should be verified against the profile before final submission.

Table 4.7: Role–Permission Matrix

Component	Administrator	Student (Portal User)
Form: Student Registration	A E V D	—
Form: Company Details	A E V D	—
Form: Internship Application	A E V D	A V (own)
Forms: Attendance and Evaluation	A E V D	—
Reports (all)	V	V (permitted)
Page: Internship Dashboard	V	—
Approvals: Pending / Completed Tasks	V; approve / reject	—

A = Add, E = Edit, V = View, D = Delete; — = no access intended.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application “Internship Management System” was created in the Zoho Creator workspace of the student’s account and is built and tested in the Development environment, as indicated by the Development badge in the application header. The left-hand menu groups the application into Student Registrations, Approvals, Company Details, Internship Applications, Internship Attendances, Internship Evaluations and Internship Dashboard (Figure 5.1).The application was created from scratch using Zoho Creator, with forms, workflows, reports, and dashboards configured according to the project requirements.

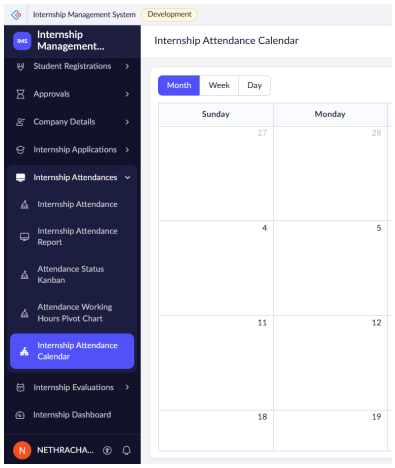


Figure 5.1: Application Navigation Sections in the Development Environment

5.2 Forms and Data Capture

Five forms were created. Student Registration (Figure 5.2) captures student details such as ID, name, e-mail, phone, department, year, CGPA, skills, resume, photo and placement interest, and also holds an Attendance Percentage field that is updated by a workflow. Company Details stores the company profile and the internship opportunity, including domain, mode, duration, stipend and available positions.

Internship Application (Figure 5.3) links a student and a company through lookup fields and records the domain, application date, start and end dates, mode, cover letter, resume, application status, approval status, the Application Approved check box and admin remarks. Internship Attendance records attendance status, check-in and check-out times, working hours, a proof document and remarks. Internship Evaluation (Figure

5.4) records the mentor name and five ratings; the Overall Rating is a formula field that averages the ratings and is therefore not entered on the form. The figures show selected forms in live mode; screenshots of the form builder were not captured.

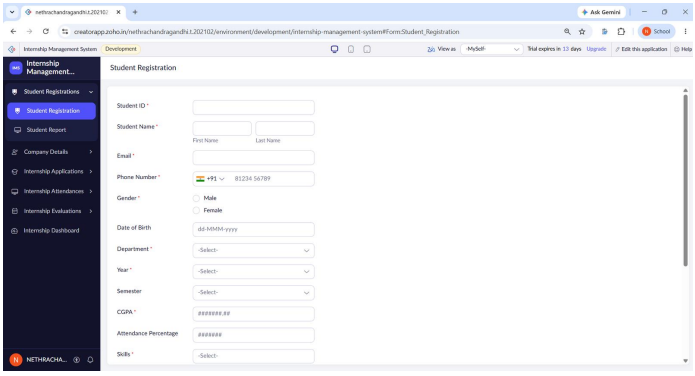


Figure 5.2: Student Registration Form in Live Mode (Top)

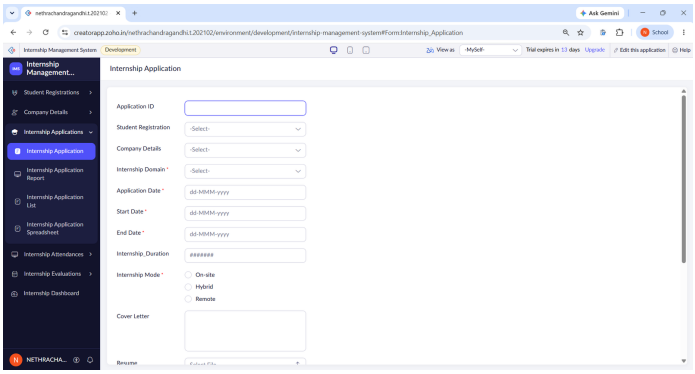


Figure 5.3: Internship Application Form in Live Mode (Top)

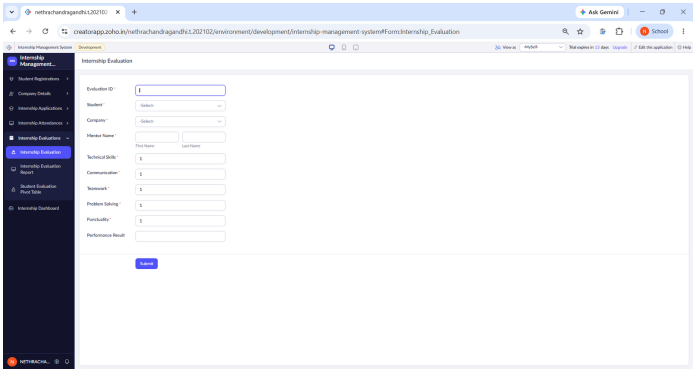


Figure 5.4: Internship Evaluation Form in Live Mode

5.3 Reports

Thirteen reports were created (Table 5.1). The figures that follow show the reports with the data stored at the time of capture.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Student Report	List	Student Registration	Default	—	Admin
Company Report	List	Company Details	Default	—	Admin
Company Location Map	Map	Company Details	Company address markers	—	Admin
Internship Application Report	List	Internship Application	Default	—	Admin
Internship Application List	List	Internship Application	Default	—	Admin
Internship Application Spreadsheet	Spreadsheet	Internship Application	Default	—	Admin
Internship Application Timeline	Timeline	Internship Application	Week and Day views	—	Admin
Internship Attendance Report	List	Internship Attendance	Default	—	Admin
Attendance Status Kanban	Kanban	Internship Attendance	Columns by Attendance Status	—	Admin
Attendance Working Hours Pivot Chart	Pivot chart	Internship Attendance	Sum of Working Hours by Attendance Date	—	Admin
Internship Attendance Calendar	Calendar	Internship Attendance	Month, Week and Day views	—	Admin
Internship Evaluation Report	List	Internship Evaluation	Default	—	Admin
Student Evaluation Pivot Table	Pivot table	Internship Evaluation	Student and Company fields	—	Admin

The Student Report (Figure 5.5) and the Company Report (Figure 5.6) list the three students and three companies stored at the time. The Company Location Map (Figure 5.7) shows company markers around Chennai. The Internship Application Report (Figure 5.8, Figure 5.9), and List (Figure 5.10) show eight applications in different layouts (the Spreadsheet layout is listed in Table 5.1); the Timeline (Figure 5.11) shows applications along a week view. Note that the Internship_Duration column is filled only for the record on which the duration workflow has run (see Section 6.4).

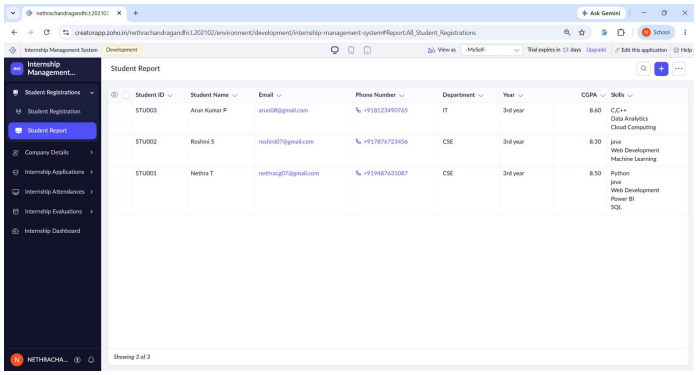


Figure 5.5: Student Report

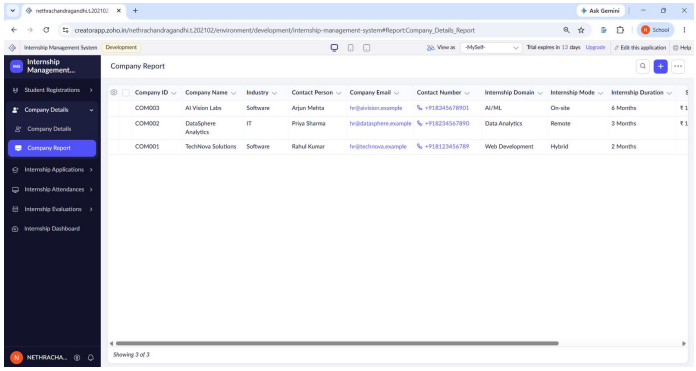


Figure 5.6: Company Report

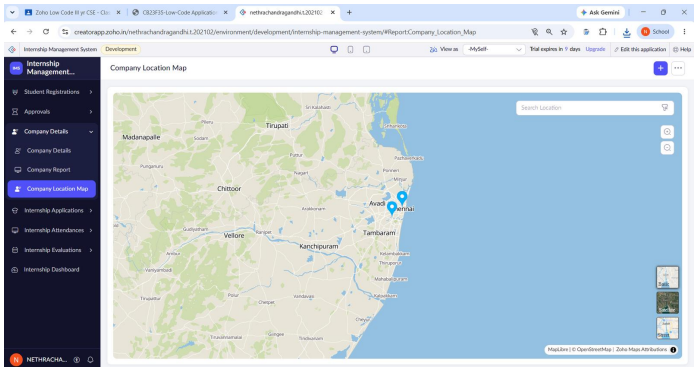


Figure 5.7: Company Location Map

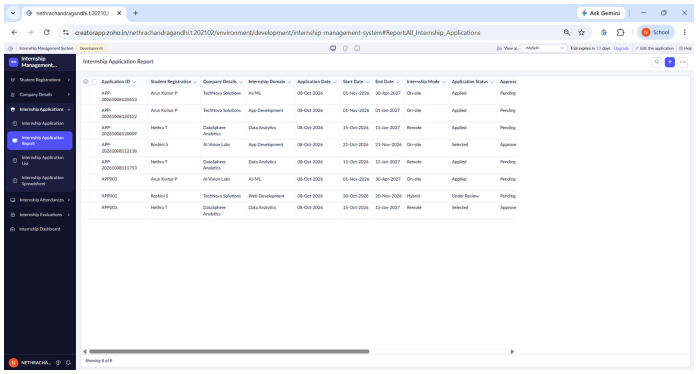


Figure 5.8: Internship Application Report (Left Columns)

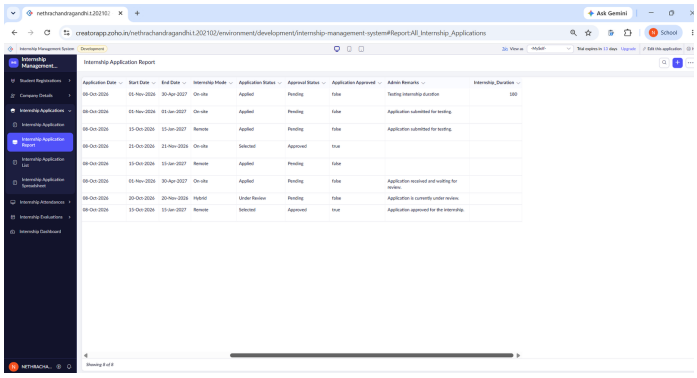


Figure 5.9: Internship Application Report (Right Columns)

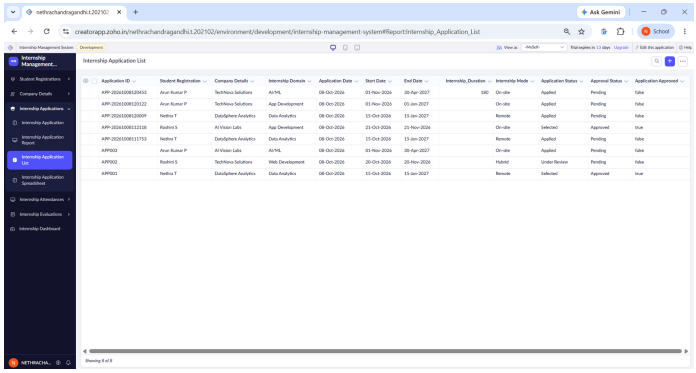


Figure 5.10: Internship Application List

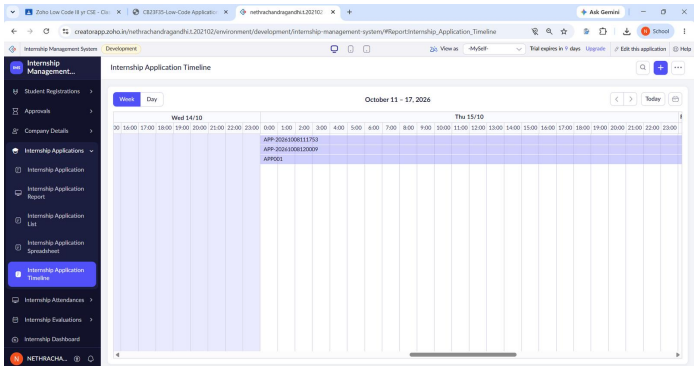


Figure 5.11: Internship Application Timeline

The Internship Attendance Report (Figure 5.12) shows four attendance records; the Attendance Status Kanban (Figure 5.13) groups them by status, with no records in the Leave and Half Day columns. The Pivot Chart plots the sum of working hours against the attendance date, and the Calendar places each record on its date (see Table 5.1). The Internship Evaluation Report (Figure 5.14) shows the five ratings and the overall rating for three evaluations.

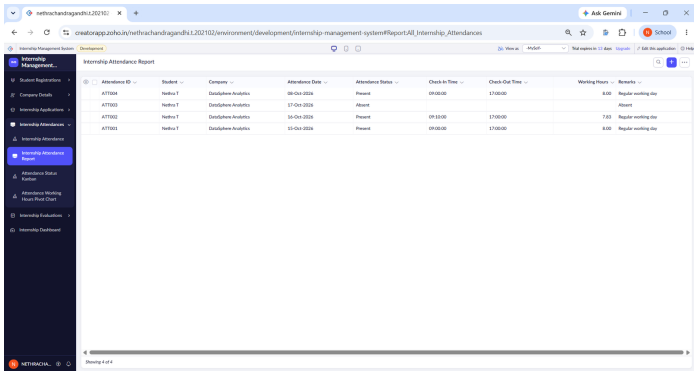


Figure 5.12: Internship Attendance Report

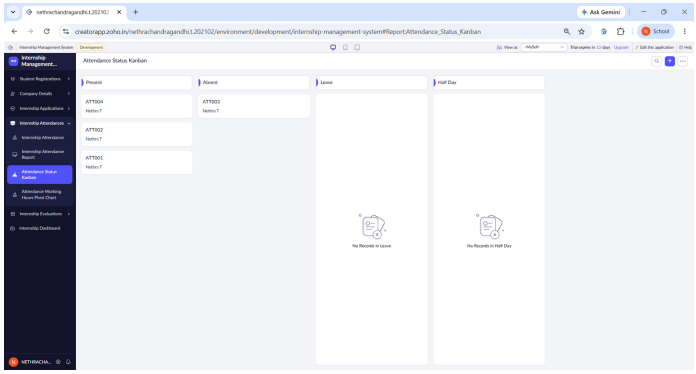


Figure 5.13: Attendance Status Kanban

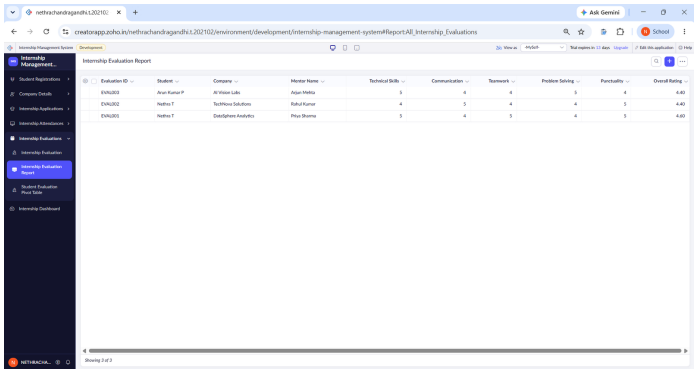


Figure 5.14: Internship Evaluation Report

5.4 Pages and Dashboards

The Internship Dashboard (Figure 5.15) gives a consolidated view. It contains a Total Students panel, a Total Companies panel, an Internship Application (total applications) panel, an Application Status pie chart, an Internship Working Hours bar chart and an Average Evaluation Rating gauge on a scale of 0 to 5. A Selected Applications panel is also part of the dashboard .The captured values were 3 students, 3 companies and 8 applications; the pie chart shows the statuses Applied, Under Review and Selected; the bar chart shows working hours for 8, 15 and 16 October 2026; and the gauge shows an average rating of 4.47. These values change as records are added.

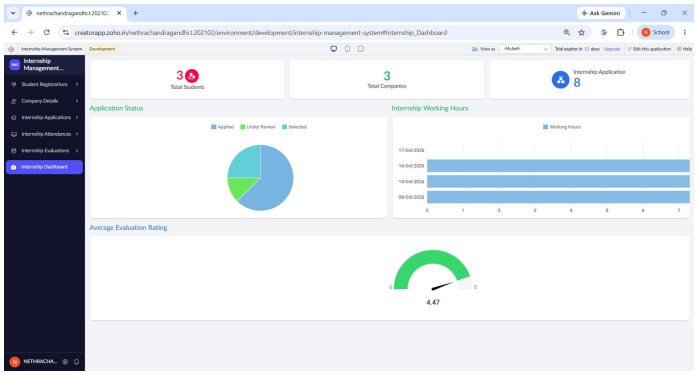


Figure 5.15: Internship Dashboard

5.5 Deluge Scripting

Seven Deluge scripts are used, all inside form workflows (Table 5.2). Six run in the "Validations on form submission" event and one runs on the update of the Application Approved field. They use conditions, date functions (daysBetween), time functions (getHour, getMinutes, getSeconds), arithmetic with rounding, string concatenation with the current time, a criteria-based fetch with a for each loop, and the sendmail task. No custom functions were written. The complete source is given in Appendix C.

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Internship Application → Created → On validate	Generate the Application ID	zoho.currenttime, toString, string concatenation
DS-02	Internship Application → Edited → On validate	Calculate internship duration in days	null check, daysBetween
DS-03	Internship Application → Edited → On validate	Set Approval Status and Application Status when approved	if condition, field assignment
DS-04	Internship Application → Edited → Update of Application Approved	Send the approval notification e-mail	sendmail
DS-05	Internship Attendance → Created → On validate	Calculate working hours from check-in and check-out	null check, getHour / getMinutes / getSeconds, round
DS-06	Internship Evaluation → Edited → On validate	Derive Performance Result from Overall Rating	if / else if
DS-07	Student Registration → Edited → On validate	Calculate attendance percentage	criteria fetch, for each loop

Code 5.1: Application ID generation (DS-01)

```
input.Application_ID = "APP-" +
zoho.currenttime.toString("yyyyMMddHHmmss");
```

Code 5.2: Attendance working hours (DS-05)

```
if(input.Check_In_Time != null && input.Check_Out_Time != null)
{
    checkin_seconds = input.Check_In_Time.getHour() * 3600 +
input.Check_In_Time.getMinutes() * 60 +
input.Check_In_Time.getSeconds();
    checkout_seconds = input.Check_Out_Time.getHour() * 3600 +
input.Check_Out_Time.getMinutes() * 60 +
input.Check_Out_Time.getSeconds();
    if(checkout_seconds >= checkin_seconds)
    {
        input.Working_Hours = ((checkout_seconds - checkin_seconds) /
3600.0).round(2);
    }
}
```

Code 5.3: Application approval status update (DS-03)

```
if(input.Application_Approved == true)
{
    input.Approval_Status = "Approved";
    input.Application_Status = "Selected";
}
```

DS-01 builds the Application ID from the prefix “APP-” and the current date and time (for example APP-20261008120453), as seen in Figure 5.8. DS-05 converts the check-in and check-out times into seconds, subtracts them and divides by 3600 to give hours rounded to two decimals; for 09:00 to 17:00 it produced 8.00 and for 09:10 to 17:00 it produced 7.83 (Figure 5.12). DS-03 sets Approval Status to Approved and Application Status to Selected when Application Approved is true. The workflow screens are shown in Section 5.6.

5.6 Workflows and Automation

Table 5.3 lists the workflows: seven form workflows, two schedules, one approval process and one Blueprint. No batch or report workflows were created.

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Application Submission Workflow	Form workflow	Internship Application created, on validate	—	Generate Application ID	DS-01

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Calculate Internship Duration	Form workflow	Internship Application edited, on validate	Start and end dates not null	Store duration in days	DS-02
Application Approval Workflow	Form workflow	Internship Application edited, on validate	Application Approved is true	Set Approved and Selected	DS-03
Student Approval Email Notification	Form workflow	Update of Application Approved	—	Send e-mail	DS-04
Attendance Working Hours Workflow	Form workflow	Internship Attendance created, on validate	Check-in and check-out not null	Calculate working hours	DS-05
Calculate Performance Result	Form workflow	Internship Evaluation edited, on validate	Overall Rating thresholds	Set Performance Result	DS-06
Calculate Attendance Percentage	Form workflow	Student Registration edited, on validate	Present records	Compute attendance percentage	DS-07
Daily Pending Application Check	Schedule	Starts 15-Oct-2026, 09:00	—	Send mail to the admin user	—
Monthly Internship Summary Reminder	Schedule	Starts 01-Nov-2026, 09:00	—	Reminder action	—
Internship Application Approval Process	Approval	Application record (Pending Tasks)	Actions run if Application_Status is “Applied”	Level 1 approver; Approve and Reject; mail to submitter	—
Internship Application Process	Blueprint	Stage transitions	—	Applied → Under Review → Selected / Rejected	—

The Application Approval Workflow is shown in Figure 5.16. The Application Submission Workflow (DS-01), Application Approval Workflow (Figure 5.16), Calculate Internship Duration (DS-02), Attendance Working Hours Workflow (DS-05), Calculate Performance Result (DS-06) and Calculate Attendance Percentage (DS-07) each contain one Deluge script (full code in Appendix C). In the Performance Result

script, ratings of 4.5 or more give “Excellent”, 3.5 or more “Good” and 2.5 or more “Average”.

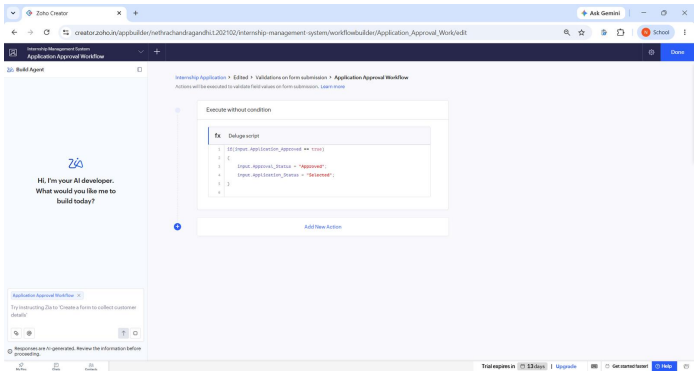


Figure 5.16: Application Approval Workflow (DS-03)

The Blueprint “Internship Application Process” (Figure 5.17) defines the stages Applied, Under Review, Selected and Rejected, with the transitions Review Application, Select Application and Reject Application. Two custom schedules are enabled (Figure 5.18); the Daily Pending Application Check sends a mail to the administrator user at 09:00.

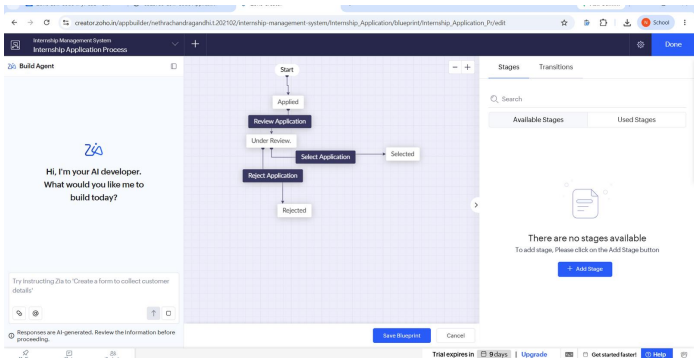


Figure 5.17: Internship Application Process Blueprint

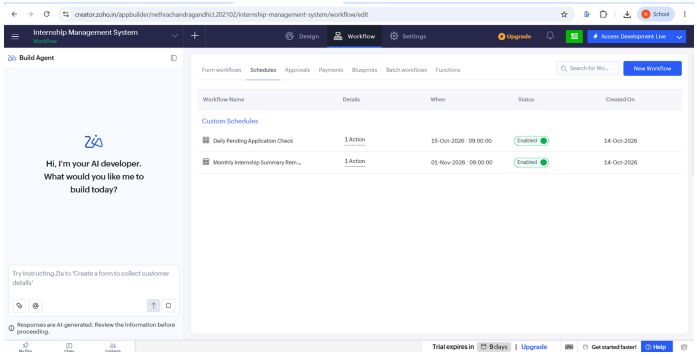


Figure 5.18: Custom Schedules

The Internship Application Approval Process (Figure 5.19) has one Level 1 approver with Approve and Reject outcomes; each outcome runs a mail action only when

Application_Status is “Applied”. A newly added application appears under Approvals → Pending Tasks (Figure 5.20). This approval process is separate from the Application Approved workflow (DS-03, DS-04), which acts on the check box in the record itself.

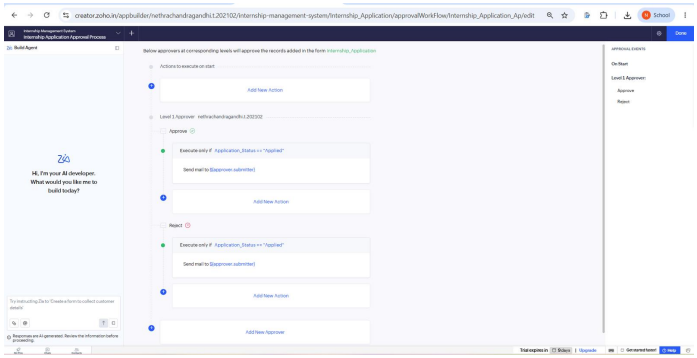


Figure 5.19: Internship Application Approval Process Configuration

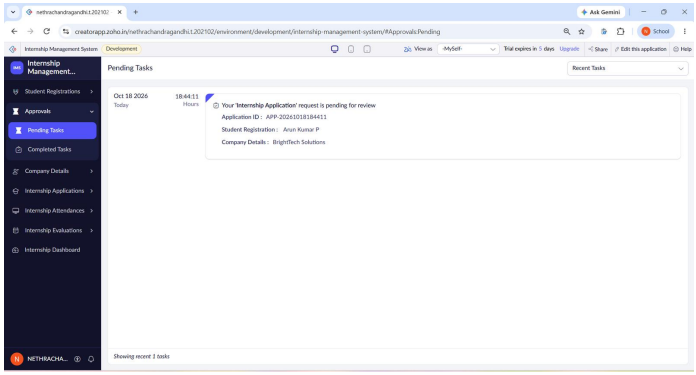


Figure 5.20: Pending Tasks in the Approvals Section

5.7 Integrations

No external systems, REST APIs, webhooks or connections were used. The application uses two built-in platform capabilities, listed in Table 5.4: the Send Mail action for the approval notification and the Map report for company locations.

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Approval e-mail	Zoho Creator Send Mail	sendmail task in DS-04	Platform-managed	Notify that the application is approved
Company map	Zoho Maps	Map report	Platform-managed	Show company locations from addresses

Code 5.4: Approval notification e-mail (DS-04)

```
sendmail
[
    from :"<configured sender address>"
    to :"<fixed test e-mail address>"
    subject :"Internship Application Approved"
    message :"Dear Student, your internship application has been
approved. Please log in to the Internship Management System to view
your application details."
]
```

The e-mail is sent with the subject “Internship Application Approved”. It was tested with a fixed test address, and the message was received (Figure 5.21). Delivery to each student’s own registered address has not been tested.

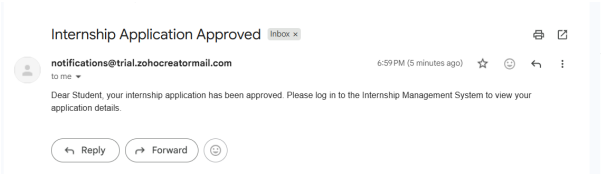


Figure 5.21: Approval Notification Received in the Inbox

5.8 User Management and Role-Based Access

The role hierarchy contains the CEO role with the Admin role beneath it (Figure 5.22). Portal user permission profiles named Customer and Student are defined; the Student profile is described as allowing students to view permitted internship information and manage their own applications (Figure 5.23). The Zoho Creator user list shows the application owner as Super Admin along with other active users.

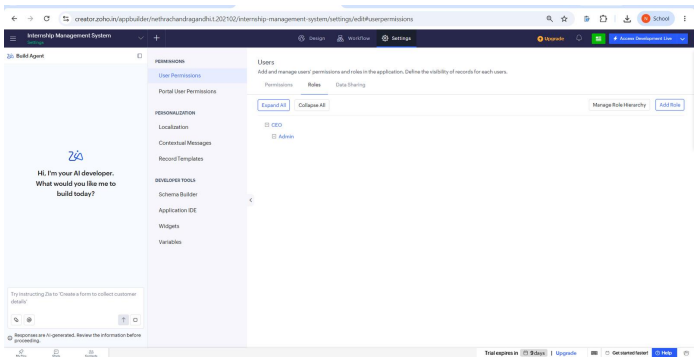


Figure 5.22: Role Hierarchy (CEO and Admin)

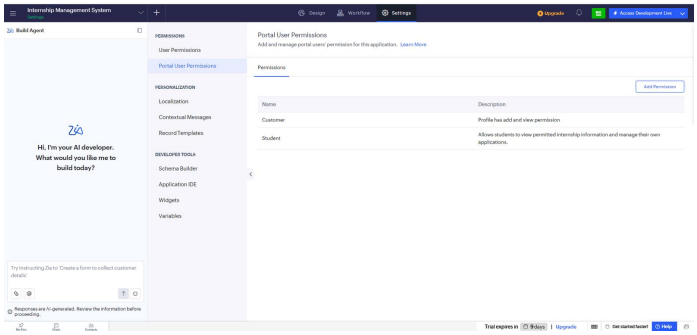


Figure 5.23: Portal User Permissions (Customer and Student)

5.9 AI and Analytics Features

Analytics are provided by the dashboard charts and pivot reports: the Application Status pie chart, the Internship Working Hours bar chart and the Average Evaluation Rating gauge (Figure 5.15), the Attendance Working Hours Pivot Chart and the Student Evaluation Pivot Table. These summarise application status, attendance hours and evaluation ratings. No Zia prediction or other AI field was configured in the forms. The Zoho Creator editor displays a Zia "Build Agent" panel in the workflow screens. The Build Agent and other AI-assisted features were not used while developing the application.

5.10 Pro-Code Extension

No pro-code extension was developed. All logic is written in Deluge inside form workflows, and the requirements did not call for custom widgets, Catalyst functions or REST API integrations. Such extensions would become relevant only if the system had to exchange data with external services or needed processing beyond what Deluge supports.

5.11 Mobile and Portal Access

Portal user permission profiles (Customer and Student) are defined in the application (Figure 5.23). A live portal login and the Zoho Creator mobile app were not demonstrated in this project.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was carried out in the Development environment by submitting records through the live forms and checking the results in the reports, the dashboard and the mailbox. The levels covered were (a) form-level testing of record submission, (b) script-level testing of each Deluge calculation, (c) workflow testing of the status update, approval request and e-mail notification, and (d) validation of the report and dashboard views. Role-based testing with a Student portal login, negative tests of mandatory fields and user acceptance testing were not recorded and are listed as pending in Section 6.5.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Submit Student Registration form	Enter valid student details	Record saved	Three students listed (Figure 5.5)	Pass
TC-02	FR-02	Submit Company Details form	Enter valid company details	Record saved	Three companies listed (Figure 5.6)	Pass
TC-03	FR-03	Submit Internship Application	Select student, company, dates	Saved; Application ID generated	ID such as APP-20261008120453 (Figure 5.8)	Pass
TC-04	FR-08	Tick Application Approved	Edit an application and tick the check box	Approval Status Approved; Application Status Selected	Record shows Selected, Approved, true (Figure 5.10)	Pass
TC-05	FR-04	Internship duration	Start 01-Nov-2026, end 01-Dec-2026	30 days	30 days (as recorded by the student); 180 days for 01-Nov-2026 to 30-Apr-2027 (Figure 5.9)	Pass
TC-06	FR-05	Working hours, full day	Check-in 09:00, check-out 15:00	6.00 hours	6.00 hours (as recorded); 09:00–17:00 gave 8.00 (Figure 5.12)	Pass
TC-07	FR-05	Working hours, late check-in	Check-in 09:10, check-out 17:00	7.83 hours	7.83 hours (Figure 5.12)	Pass
TC-08	FR-05	Absent record	Status Absent, no times	No working hours	Working Hours blank for ATT003 (Figure 5.12)	Pass
TC-09	FR-06	Overall rating, example 1	5, 4, 5, 4, 5	4.6	4.60 (Figure 5.14)	Pass
TC-10	FR-06	Overall rating, example 2	5, 4, 4, 5, 4	4.4	4.40 (Figure 5.14)	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-11	FR-09	Approval e-mail	Tick Application Approved	E-mail sent to the configured test address	E-mail received (Figure 5.21)	Pass
TC-12	FR-10	Approval request raised	Add an application	Pending task for the approver	Pending task APP-20261018184411 (Figure 5.20)	Pass
TC-13	FR-13	Attendance Kanban	Open the Kanban report	Records grouped by status	Present 3, Absent 1; Leave and Half Day empty (Figure 5.13)	Pass
TC-14	FR-14	Dashboard values	Open the dashboard	Counts, charts and gauge shown	3 students, 3 companies, 8 applications, gauge 4.47 (Figure 5.15)	Pass
TC-15	FR-11	Blueprint transitions	Move an application through the stages	Stage changes per transition	Blueprint transition completed successfully (Figure 5.17).	Pass
TC-16	FR-15	Scheduled mail	Wait for the schedule time	Mail sent to admin	Scheduled email executed successfully (Figure 5.18).	Pass

6.3 Debugging Deluge Scripts

Each script was verified by submitting or editing a record and reading the calculated field in the report. For example, the working-hours script was checked with known check-in and check-out times and the duration script with known dates. Where a result was missing, the trigger event of the workflow (create or edit) and the field link names used in the script were checked in the Workflow builder. Debug output or log screenshots were not captured, so no figure is included for this section.

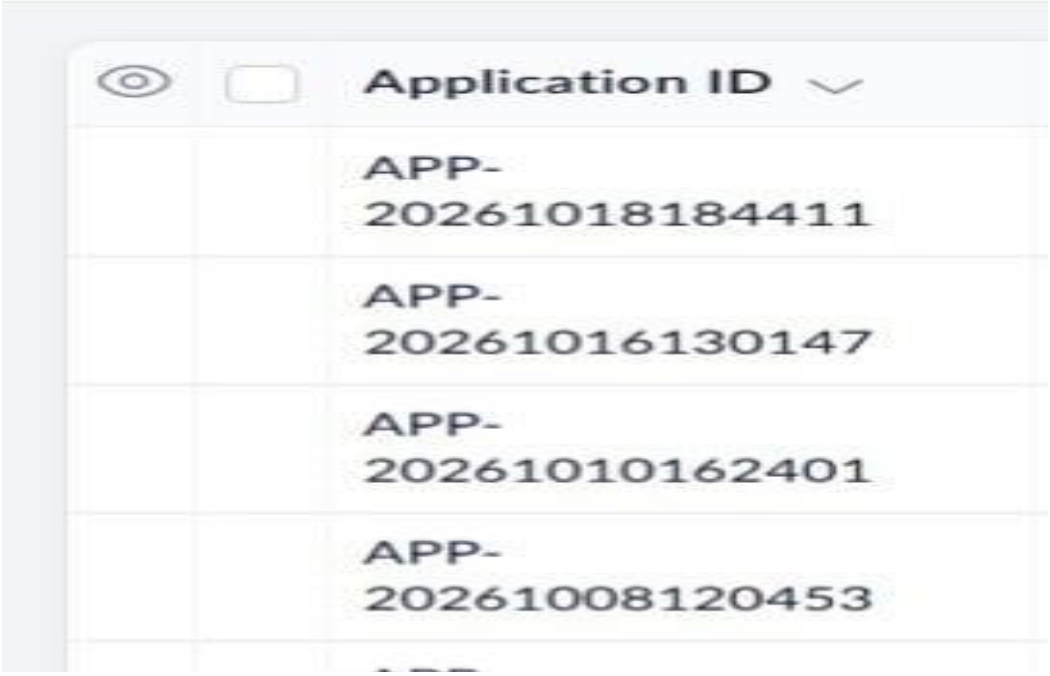


Figure 6.1: Deluge Debug Output

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Internship_Duration is blank for most applications (Figure 5.10).	The duration workflow is configured for the Edit event only, so records that are only created do not run it.	Not yet applied; run the workflow on create as well as edit.	Open
BUG-02	The approval e-mail is not addressed to the student concerned.	The recipient address in DS-04 is a fixed test address.	Not yet applied; use the student's registered e-mail address through the lookup.	Open

6.5 User Acceptance Testing

User acceptance feedback was collected from one invited user who accessed the application with read and write permissions. The user reviewed the application and provided positive feedback on its usability. The feedback is recorded in Table 6.3.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
User 1	Application user (Read/Write)	The application is easy to use and the features are good.	Feedback received; no changes requested.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was developed and tested in the Development environment of Zoho Creator, which is indicated by the Development badge and the "Access Development Live" option in the screenshots. The account is on a trial edition. The application was developed and tested in the Development environment. Although Stage and Production environments were available in the Zoho Creator Environments section, the application was not separately published to Production. Therefore, the application was maintained and tested in Development.

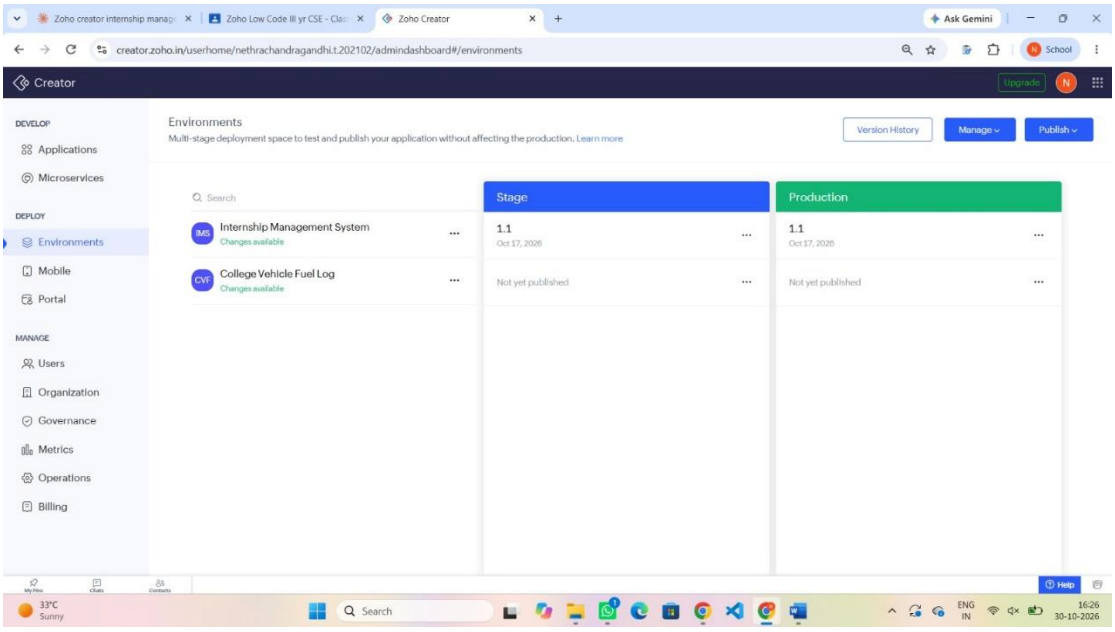


Figure 7.1: Environment/Publish to Production

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	Partly (see Table 6.1)
2	Sample/test data removed or isolated	✓ Test data retained for demonstration
3	Roles, permissions and users configured in Production	Not Applicable
4	Connections authorised in Production	Not applicable
5	Schedules and workflows enabled	✓ (Development)
6	Application published and URL verified	Not published seperately
7	Version / release notes recorded	✓ Version 1.1 recorded

7.2 Performance Optimisation

No formal performance measurements were taken, because the data volume is small. One design choice is relevant to performance: DS-07 retrieves only the attendance records of the student concerned through a criteria-based fetch instead of reading the whole form.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Criteria-based fetch of one student's attendance records	DS-07	Not measured	Not measured	Not measured

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role hierarchy (CEO and Admin)	Application users	Unauthorised administrative actions
Portal user permission profiles (Customer, Student)	Portal users	Students accessing records meant for others
Approval process with a designated Level 1 approver	Internship Application	Unreviewed approval of applications
Mandatory fields	All forms	Incomplete or invalid records

Encryption of individual fields, OAuth connections and record-level permission criteria were not configured, because the application has no external connections and no field-level sensitivity requirement was defined.

7.4 Governance and Administration

Administration is handled by the application owner, who is the Super Admin of the Zoho Creator account and the Level 1 approver of the approval process. Workflows are given descriptive names (for example, Calculate Internship Duration) so that each rule can be located from the Workflow builder, and the user list is maintained in the Zoho Creator administration console. Version information was reviewed through the Zoho Creator Environments section, where Version 1.1 was displayed.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The following scenario demonstrates the working of the Internship Management System developed using Zoho Creator.

Step 1: Student Registration and Application Submission

The student details are entered and stored through the Student Registration form. The student then submits an internship application by selecting the company and providing the required internship details. The system automatically generates a unique Application ID for tracking the application.

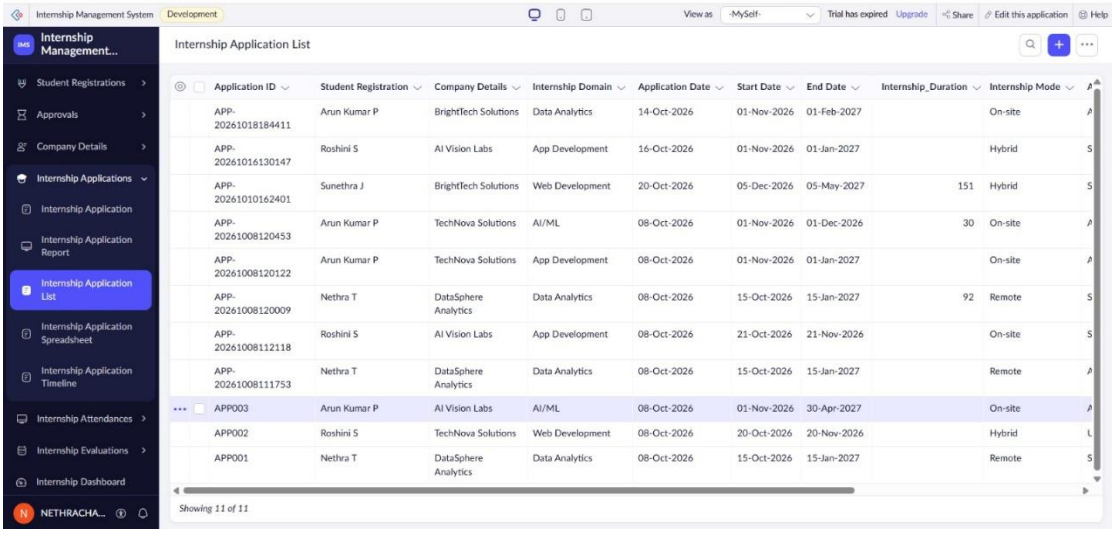


Figure 8.1: Student Registration and Internship Application Submission

Step 2: Application Approval Process

After the application is submitted, an approval request appears under the Pending Tasks section. The administrator reviews the student's application and verifies the submitted internship details before making the approval decision. Once approved, the application status is updated accordingly.

The approval workflow automatically updates the Approval Status to Approved and the Application Status to Selected when the application is approved. This helps simplify the review process and maintain accurate application records.

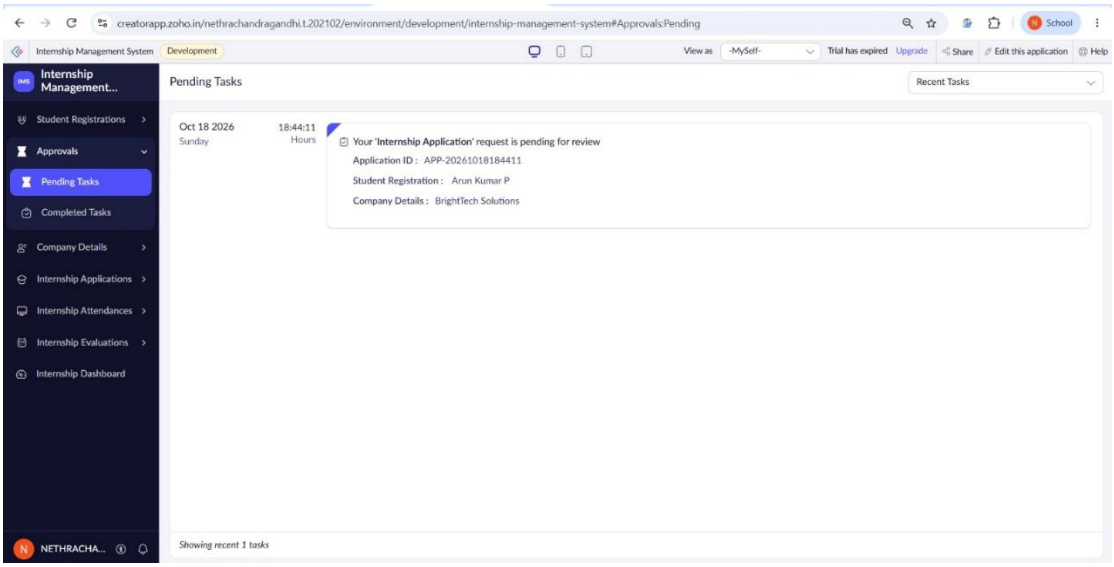


Figure 8.2: Pending Internship Application for Administrator Review

Step 3: Dashboard and Application Monitoring

The Internship Dashboard provides an overview of the system, including the total number of students, companies and internship applications. It also displays the application status distribution, internship working hours and average evaluation rating, helping the administrator monitor internship activities.

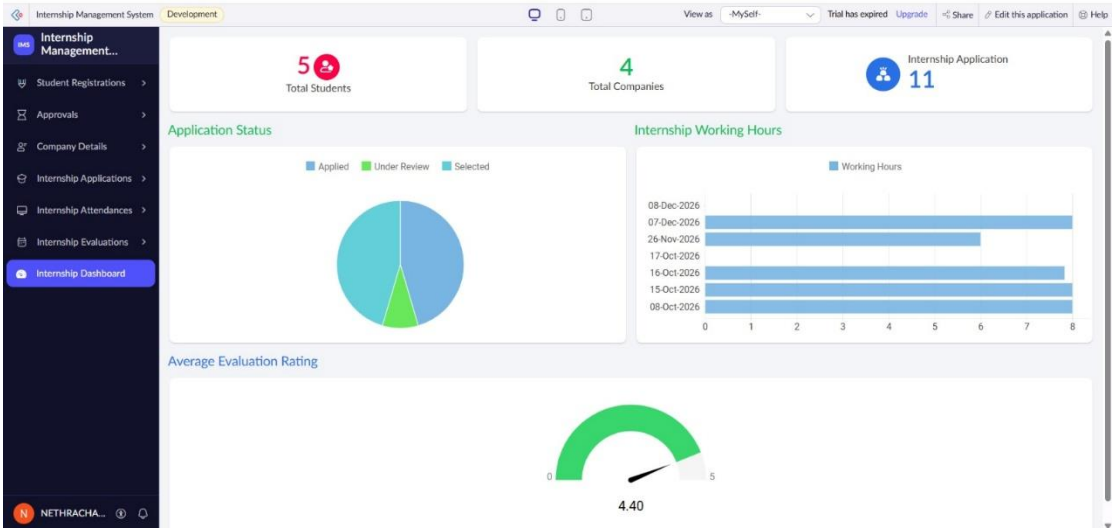


Figure 8.3: Internship Management Dashboard

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
1. Forms and linked data model	Section 4.2, 4.3, 5.2	Five forms with lookup fields (Figure 4.2, Figure 5.2–Figure 5.4)
2. Deluge automation	Section 5.5, 5.6	Seven scripts; results in Figure 5.8, Figure 5.12 and Figure 5.14
3. Application life cycle and approval	Section 5.6, 5.7	Approval process, Blueprint, status workflow and e-mail (Figure 5.19, Figure 5.17, Figure 5.21)
4. Reports and dashboard	Section 5.3, 5.4	Thirteen reports and one dashboard (Table 5.1, Figure 5.15)
5. Role-based access	Section 5.8	Role hierarchy and portal profiles (Figure 5.22, Figure 5.23)
6. Testing	Chapter 6	Fourteen test cases passed; two pending (Table 6.1)

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Application ID	Assigned or typed manually	Generated by DS-01	Removes manual numbering
Duration and working hours	Calculated by hand	Calculated by DS-02 and DS-05	Fewer calculation errors
Status tracking	Informal follow-up	Approval process, Blueprint and status fields	Defined stages
Report generation	Compiled from separate files	Thirteen ready reports and a dashboard	Consolidated view
Notification	Informed manually	E-mail action on approval (fixed test address tested)	Automated, with dynamic delivery pending

No timing measurements were taken, so the improvements are stated qualitatively.

8.4 Limitations of the Low-Code Approach

The application depends on the features and limits of the Zoho Creator platform and was built on a trial edition, so continued use needs a subscription. Some logic is limited by the trigger the platform offers: the duration workflow runs on edit, which leaves

most records without a duration (BUG-01). The notification uses a fixed address, and role-based testing has not been carried out. These points can be addressed within the platform (Section 9.2); more demanding needs, such as external integrations, would require Deluge connections or a pro-code extension.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of the application sits at the no-code end: the five forms, thirteen reports, dashboard, approval process, Blueprint, schedules, roles and portal profiles are configured visually. The seven Deluge scripts place the calculations and the e-mail action at the low-code level. No pro-code component was needed. This split kept the development effort on the internship process and left hosting and security to the platform.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Internship Management System was developed on Zoho Creator to manage internship activities through forms, workflows, reports and a dashboard. Its five forms cover student registration, company information, internship applications, attendance and evaluation. Deluge workflows generate application IDs, calculate internship duration, working hours, performance result and attendance percentage, and update application status. An approval process, a Blueprint and two schedules support the application life cycle.

Testing in the Development environment showed that the calculations, status updates, approval request, reports and dashboard work as intended, and that the approval e-mail was delivered to the configured test address. The pending items are dynamic e-mail delivery to students, role-based testing, user acceptance feedback and deployment beyond Development. The project shows that a low-code platform can deliver a working internship record-management solution with limited scripting.

9.2 Future Enhancements

The following are proposed enhancements and are not part of the completed work:

- Send approval notifications to each student's registered e-mail address.
- Run the duration and attendance-percentage workflows when a record is created as well as when it is edited.
- Implement more detailed user-role access and permissions.
- Add further approval stages where required.
- Improve report filters and analytical summaries.
- Add certificate management if required by a future scope, and extend the system with additional notifications and analytics.

9.3 Learning Outcomes and Career Relevance

The project gave practical experience in data modelling with forms and lookups, Deluge scripting, workflow automation, approvals, Blueprint design, reports and dashboards, and in testing a low-code application. These skills are relevant to roles such as low-

code developer and business-application developer, and the project shows how a citizen developer can progress towards professional development by adding scripting and, later, pro-code extensions.

REFERENCES

[1] W. Rosam and O. Kalnichevski, *Low-Code Development: A Practical Guide*. New Delhi, India: BPB Publications.

[2] Zoho Corporation, "Zoho Creator Help Documentation." [Online]. Available: https://www.zoho.com/creator/help/ (Accessed: 15 October 2026).

[3] Zoho Corporation, "Deluge Script Reference Guide." [Online]. Available: https://www.zoho.com/deluge/ (Accessed: 18 October 2026).

[4] Zoho Corporation, "Zoho Creator Learning Centre." [Online]. Available: https://www.zoho.com/creator/resources/ (Accessed: 15 October 2026).

[5] Zoho Corporation, "Zoho Creator API v2 Documentation." [Online]. Available: https://www.zoho.com/creator/help/api/v2/ (Accessed: 21 October 2026).

[6] Zoho Corporation, "Zoho Creator: Low-Code Application Development Platform." [Online]. Available: https://www.zoho.com/creator/ (Accessed: 21 October 2026).

[7] C. Richardson and J. R. Rymer, "New Development Platforms Emerge for Customer-Facing Applications," Forrester Research, Cambridge, MA, USA, Jun. 2014.

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Y = implemented; Partial = partly implemented, with the gap stated; N = not implemented in this project.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Internship Management System with seven menu sections	5.1	Figure 5.1
II	Forms with ≥ 8 distinct field types	Y	Five forms	5.2	Figure 5.2–Figure 5.4
II	Lookup field (relationship between forms)	Y	Application, Attendance, Evaluation	4.2, 5.2	Figure 4.2, Figure 5.3
II	Subform (parent–child data)	Y	used	4.2, 5.2	—
II	Field validations / layout rules	Y	Mandatory fields on all forms; rating range 1 to 5	5.2	Figure 5.2, Figure 5.4
II	List report with filter, grouping, custom action	Y	List reports created; filters, grouping and custom actions not configured	5.3	Figure 5.5, Figure 5.10
II	Kanban / Calendar / Timeline / Map report	Y	All four created	5.3	Figure 5.13, Table 5.1, Figure 5.11, Figure 5.7
II	Pivot table / chart report	Y	Pivot chart and pivot table	5.3	Table 5.1
II	Page / dashboard built with UI Builder	Y	Internship Dashboard	5.4	Figure 5.15
II	Deluge – form events (validate / success / field actions)	Y	Validate events and field-update event	5.5	Code 5.1–Code 5.3
II	Deluge – functions, fetch, loops, collections	Y	Fetch and for each in DS-07; no custom functions	5.5	Appendix C
II	Email / in-app notifications	Y	sendmail in DS-04; schedule mail	5.5, 5.6	Code 5.4, Figure 5.21
II	Form workflow	Y	Seven form workflows	5.6	Figure 5.16
II	Scheduled workflow	Y	Two schedules	5.6	Figure 5.18
II	Approval process	Y	Internship Application Approval Process	5.6	Figure 5.19, Figure 5.20
II	Blueprint	Y	Internship Application Process	5.6	Figure 5.17
II	Batch / report workflow	N	Not used	5.6	—
II	Testing & debugging evidence	Y	Sixteen test cases, two pending	6	Table 6.1

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Integration with a Zoho service	N	Only built-in Send Mail and Map report	5.7	—
III	Connection / REST API call (invokeurl)	N	Not used	5.7	—
III	Webhook or Creator REST API	N	Not used	5.7	—
III	Users, roles and permission profiles	Y	CEO and Admin roles; Customer and Student profiles	5.8	Figure 5.22, Figure 5.23
III	Record-level access criteria / customer portal	Y	Portal profiles defined; record-level criteria not verified	5.8	Figure 5.23
III	Zia / AI feature	N	Zia Build Agent panel displayed; AI-assisted features not used	5.9	—
III	Analytics (charts / Zoho Analytics)	Y	Dashboard charts, gauge and pivot reports	5.9	Figure 5.15, Table 5.1
IV	Performance optimisation applied & measured	N	Not measured	7.2	—
IV	Deployment across environments	N	Development only	7.1	—
IV	Security controls (encryption, OAuth, permissions)	Y	Permissions and approval only	7.3	Table 7.3
IV	Governance (audit trail, administration)	N	Not evidenced	7.4	—
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not used	5.10	—
V	Low-code limitations & spectrum analysis	Y	Discussed	8.4, 8.5	—
—	Mobile app / portal access	Y	Portal profiles defined; mobile and live portal not demonstrated	5.11	Figure 5.23

Total features implemented: 23/ 32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	4	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	4	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	2	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	3	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	3	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Application Submission Workflow

```
// Script ID : DS-01
// Location  : Internship Application → Created → Validations on form
submission
// Purpose   : Generate the Application ID

input.Application_ID = "APP-" +
zoho.currenttime.toString("yyyyMMddHHmmss");
```

C.2 Calculate Internship Duration

```
// Script ID : DS-02
// Location  : Internship Application → Edited → Validations on form
submission
// Purpose   : Calculate the internship duration in days

if(input.Start_Date != null && input.End_Date != null)
{
    duration = input.Start_Date.daysBetween(input.End_Date);
    input.Internship_Duration = duration;
}
```

C.3 Application Approval Workflow

```
// Script ID : DS-03
// Location  : Internship Application → Edited → Validations on form
submission
// Purpose   : Update approval and application status

if(input.Application_Approved == true)
{
    input.Approval_Status = "Approved";
    input.Application_Status = "Selected";
}
```

C.4 Student Approval Email Notification

```
// Script ID : DS-04
// Location  : Internship Application → Edited → Update of Application
Approved
// Purpose   : Send the approval notification e-mail

sendmail
[
    from :"<configured sender address>"
    to :"<fixed test e-mail address>"
    subject :"Internship Application Approved"
    message :"Dear Student, your internship application has been
approved. Please log in to the Internship Management System to view
your application details."
]
```

C.5 Attendance Working Hours Workflow

```
// Script ID : DS-05
// Location  : Internship Attendance → Created → Validations on form
submission
// Purpose   : Calculate working hours

if(input.Check_In_Time != null && input.Check_Out_Time != null)
{
    checkin_seconds = input.Check_In_Time.getHour() * 3600 +
input.Check_In_Time.getMinutes() * 60 +
input.Check_In_Time.getSeconds();
    checkout_seconds = input.Check_Out_Time.getHour() * 3600 +
input.Check_Out_Time.getMinutes() * 60 +
input.Check_Out_Time.getSeconds();
    if(checkout_seconds >= checkin_seconds)
    {
        input.Working_Hours = ((checkout_seconds - checkin_seconds) /
3600.0).round(2);
    }
}
```

C.6 Calculate Performance Result

```
// Script ID : DS-06
// Location  : Internship Evaluation → Edited → Validations on form
submission
// Purpose   : Derive Performance Result from Overall Rating

if(input.Overall_Rating >= 4.5)
{
    input.Performance_Result = "Excellent";
}
else if(input.Overall_Rating >= 3.5)
{
    input.Performance_Result = "Good";
}
else if(input.Overall_Rating >= 2.5)
{
    input.Performance_Result = "Average";
}
else
{
    input.Performance_Result =”Poor”;
}
```

C.7 Calculate Attendance Percentage

```
// Script ID : DS-07
// Location  : Student Registration → Edited → Validations on form
submission
// Purpose   : Calculate attendance percentage

attendance_records = Internship_Attendance[Student == input.ID];
total_days = 0;
attended_days = 0.0;
for each  record in attendance_records
{
    if(record.Attendance_Status == "Present")
    {
        total_days = total_days + 1;
        attended_days = attended_days + 1;
    }
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the application link in a web browser (the Development URL is given in the Project at a Glance table) and sign in with the Zoho account that has access. The menu on the left lists the modules.

D.2 Instructions for the Administrator

- Open Student Registrations → Student Registration, fill the mandatory fields and click Submit.
- Open Company Details → Company Details, enter the company and internship details and click Submit.
- Open Internship Applications → Internship Application, select the student and company, enter the domain, dates and mode, and click Submit. The Application ID is generated automatically.
- Open Approvals → Pending Tasks to see applications awaiting review and approve or reject them.
- To mark an application approved, edit the record and tick Application Approved; the status fields update and the notification e-mail is sent.
- Open Internship Attendances → Internship Attendance, enter the status and the check-in and check-out times, and click Submit; working hours are calculated.
- Open Internship Evaluations → Internship Evaluation, enter the mentor name and the five ratings, and click Submit; the overall rating is calculated.
- Use the reports under each module and the Internship Dashboard to view records and summaries.

D.3 Instructions for the Student (Portal User)

- Sign in to the portal with the credentials supplied by the administrator.
- View the internship information that is permitted for the Student profile.
- Submit and manage your own internship applications.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	NETHRACHANDRAGANDHI T	Register No.	240701357
Project Title	Internship Management System using Zoho Creator	Date	11.11.2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	NETHRA CHANDRAGANDHI T	Register No.	240701357
Department / Section	CSE	Date of Viva	11.11.2026
Application URL Verified	YES	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER

Name: _____

EXTERNAL EXAMINER

Name: _____